

INSIDE JEB

Stressed chickadees get hot under the collar to save energy



Feeling stressed can make some animals' blood run cold, while others feel hot under the collar and Joshua Robertson, Gabriela Mastromonaco and Gary Burness, from Trent University and Toronto Zoo, Canada, explain that 'the functional value of this phenomenon is poorly understood'. One possibility was that a drop in surface temperature could be attributed to terrified creatures shunting blood away from the skin to minimise blood loss in the event of an injury; although it wasn't clear why other animals run warm. However, the Canadian team had a hunch that fluctuating skin temperatures could provide a mechanism for saving energy. Robertson says, 'Birds usually dissipate heat by evaporative cooling, which is more energetically- and resource-demanding than dry heat dissipation'. The researchers wondered if stressed birds avoid wasting the energy that they would otherwise need to expend panting by

dumping the heat generated by their anxiety through their hot skins instead. In addition, the trio suspected that birds may shunt blood flow away from the skin when air temperatures turn cold, to capitalise on the heat generated by the birds' state of anxiety to keep warm at no extra cost.

To test the idea, the trio filmed black-capped chickadees with a temperature-sensing camera under a range of stressful situations – ranging from being harassed by a stuffed hawk to the presence of humans – from late spring to early summer to find out how the temperature of the patch of skin around their eyes varied. And when the team compared the chickadees' temperature variation with the air temperature, they found that the stressed birds' skin temperatures rose in sizzling conditions when they would otherwise have been expending energy in order to remain cool. And when the

conditions were chilly, the anxious birds diverted blood away from their surface, cooling the skin around their eyes to conserve the warmth produced by the tense situation and save energy.

The stressed chickadees seemed to be adjusting the blood flow to their skin to reduce their energy consumption depending on the prevailing weather, and Robertson adds, 'it does seem risky to enhance perfusion of blood to the skin during stress exposure, but our results suggest that perhaps it is a risk chickadees are willing to take'.

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